

HYDRON VIRTUAL ASSISTANT

Detailed Guide: Working of 6 Core Functions & Technical Architecture

Function 1: time / date (System Clock Query)

Command:	time OR date
Class & File:	TimeTask (TimeTask.h, TimeTask.cpp)
OOP Concept:	Inheritance & Dynamic Polymorphism
How It Works:	Jab user 'time' ya 'date' type karta hai, CommandParser TimeTask ka instance return karta hai. TimeTask::execute() C++ std::chrono::system_clock::now() aur std::localtime() se system time fetch karta hai. Formatted output (YYYY-MM-DD HH:MM:SS) console par print karta hai.

Function 2: open (Desktop App Launcher)

Command:	open notepad / open calc / open safari
Class & File:	OpenAppTask (OpenAppTask.h, OpenAppTask.cpp)
OOP Concept:	Inheritance, Polymorphism, Platform Abstraction
How It Works:	User se application ka naam (argument) leta hai. Windows par std::system("start ") execute karta hai. Mac OS par smart aliases use karta hai (notepad -> TextEdit, calc -> Calculator) aur open -a command se application ko background me launch karta hai.

Function 3: search (Web Search Automation)

Command:	search C++ OOP concepts
Class & File:	SearchTask (SearchTask.h, SearchTask.cpp)
OOP Concept:	Inheritance, Encapsulation, Helper Encoding
How It Works:	Search query ke spaces ko urlencodeQuery() helper function se '+' me convert karta hai. Phir formatted URL (https://www.google.com/search?q=...) banakar system default web browser me launch kar deta hai.

Function 4: new tab (Keyboard Shortcut Automation)

Command:	new tab
Class & File:	NewTabTask (NewTabTask.h, NewTabTask.cpp)
OOP Concept:	Inheritance & Win32 API Integration
How It Works:	Windows Win32 SendInput() API ka use karke 4 keyboard input structures ka array inject karta hai: (1) VK_CONTROL press, (2) 'T' key press, (3) 'T' key release, (4) VK_CONTROL release. Isse browser me Ctrl+T new tab shortcut simulate hota hai.

Function 5: history (Command Log Persistence)

Command:	history
Class & File:	HistoryTask & Logger (Logger.h, Logger.cpp)
OOP Concept:	File Handling (I/O Streams) & Encapsulation
How It Works:	Har executed command Logger class dwara [YYYY-MM-DD HH:MM:SS] timestamp ke sath history.txt me std::ofstream (std::ios::app mode) se log hota hai. 'history' command par HistoryTask Logger::printHistory() call karke std::ifstream se poori history display karta hai.

Function 6: exit / quit & help (Lifecycle & Assistance)

Command:	exit / quit / help
Class & File:	Assistant (Assistant.h, Assistant.cpp)
OOP Concept:	Encapsulation & REPL Control Loop
How It Works:	Assistant::run() ke main while-loop ko control karta hai. 'help' par saare supported commands ki menu list render hoti hai. 'exit' milne par isRunning boolean false ho jata hai aur farewell message dekar program cleanly exit karta hai.

Summary Architecture Flow (Execution Pipeline)

User Input → Assistant::run() → Logger::logCommand() (appends to history.txt) → CommandParser::parse() (validates & creates Task*) → task->execute() (Polymorphic call) → Output rendered.